

A Level Mechanics

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Chapter 6: Forces, Friction and Inclined Planes

Why this chapter matters

Every chapter so far has dealt with forces acting horizontally or vertically. Real objects rarely sit that conveniently: forces are applied at angles, surfaces resist sliding, and planes are tilted. This chapter builds the toolkit for handling all three — resolving a force in any direction, modelling friction with a coefficient of friction μ , and analysing bodies on inclined planes — and then combines them with the connected-particle and pulley ideas from Chapter 3 into the multi-step problems that are the hallmark of A Level mechanics papers. The systematic habit of resolving perpendicular to a surface before resolving parallel to it, and the routine combination of friction, inclines and pulleys within a single question, are the skills most worth mastering here.

A quick note on the labels you will see next to each working step in this chapter, such as **M1** or **A1**: these are mark-scheme style labels borrowed from how exam answers are graded. **M1** marks a *method* mark (you get credit for setting up the right equation, even before it is solved), and **A1** marks an *accuracy* mark (you get credit once a value is correctly worked out). Seeing them attached to each step is meant to show you exactly where the marks in a real exam answer come from — not just what the final answer is.

6.1 Resolving Forces

Concept

A single force F acting at an angle θ to a chosen axis can be split into two **components**, at right angles to each other, that together have the same effect as the original force. This process is called **resolving** the force.

Why does this work? Because force is a vector: it has both a size and a direction. Any vector can be rebuilt exactly from its horizontal and vertical "shadows" (or, more generally, its shadows along any two perpendicular directions), so replacing one force with its two components changes nothing about the physics — it just makes the algebra easier to handle.

Choosing **sensible axes** — directions in which as many forces as possible lie entirely along one axis — turns a two-dimensional problem into two independent one-dimensional equations. For a particle in equilibrium on a flat surface, horizontal and vertical axes are usually sensible. Once an incline is involved (Section 6.3), axes parallel and perpendicular to the slope become the sensible choice instead, because that is the direction the normal reaction and any motion naturally line up with.

Key vocabulary

Term	Meaning
Resultant force	The single force equivalent to the combined effect of all forces acting on a particle

Component	The part of a force acting along one chosen axis (its "shadow" on that direction)
Resolve	To split a force into components along two perpendicular directions
Equilibrium	A particle is in equilibrium when the resultant force on it is zero (it is stationary or moving at constant velocity)
Sensible axes	Perpendicular directions chosen so that the resulting equations are as simple as possible

Key formulas

KEY FORMULA

$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$

where θ is measured between the force F and the axis along which the component $F \cos \theta$ acts. For a particle in equilibrium, the components in each chosen direction must separately sum to zero:

$$\Sigma F_x = 0, \quad \Sigma F_y = 0$$

Procedure

1. Draw a clear force diagram showing every force acting on the particle, with angles measured from a stated reference direction.
2. Choose two perpendicular axes (usually horizontal/vertical, unless a slope is involved) and state the positive direction chosen for each.
3. Resolve every force along each axis in turn, writing $F \cos \theta$ or $F \sin \theta$ as appropriate. A useful check: a force resolved along its own line of action contributes its full magnitude, and along the perpendicular direction contributes zero.
4. Set the sum of components in each direction equal to zero (equilibrium) or to ma (Newton's second law), and solve the resulting equation(s), simultaneously if there are two unknowns.

Worked example

EXAMPLE 6.1

Question. A particle of weight 50 N hangs in equilibrium, held by two light inextensible strings inclined at 30° and 60° to the horizontal, attached to two fixed points at the same height. Find the tension in each string.

Step 1 — resolve horizontally. Take rightward as positive. **M1**

$$T_1 \cos 30^\circ = T_2 \cos 60^\circ \quad \mathbf{A1}$$

Step 2 — resolve vertically. Take upward as positive. **M1**

$$T_1 \sin 30^\circ + T_2 \sin 60^\circ = 50 \quad \mathbf{A1}$$

Step 3 — rearrange the horizontal equation to make T_2 the subject. Using $\cos 30^\circ = \sqrt{3}/2$

and $\cos 60^\circ = 1/2$:

$$T_1(\sqrt{3}/2) = T_2(1/2) \Rightarrow T_2 = \sqrt{3} T_1$$

Step 4 — substitute $T_2 = \sqrt{3} T_1$ into the vertical equation. Using $\sin 30^\circ = 1/2$ and $\sin 60^\circ = \sqrt{3}/2$: **M1**

$$T_1(1/2) + (\sqrt{3} T_1)(\sqrt{3}/2) = 50$$

Step 5 — simplify and solve for T_1 .

$$0.5T_1 + 1.5T_1 = 50 \Rightarrow 2T_1 = 50 \Rightarrow T_1 = 25 \quad \text{A1}$$

Step 6 — substitute back to find T_2 .

$$T_2 = \sqrt{3} T_1 = \sqrt{3} \times 25 = 25\sqrt{3} \approx 43.3 \quad \text{A1}$$

Answer: $T_1 = 25$ N (string at 30°), $T_2 = 25\sqrt{3} \approx 43.3$ N (string at 60°).

Exam tip

EXAM TIP

Why this is hard: most people are comfortable resolving a single applied force, but find it harder to resolve every force in a diagram — including tensions and reactions — about a self-chosen pair of axes, and then to solve the resulting simultaneous equations. Build this skill by always using clear $R(\rightarrow)$ and $R(\uparrow)$ style labelling for each equation: mark schemes award a method mark for a correctly resolved equation independently of whether the final numerical answer is right, so a labelled but wrong-value equation can still pick up marks a correct-but-unlabelled one might not.

Common exam mistakes

- Resolving a force using $\sin \theta$ where $\cos \theta$ is needed (or vice versa) because the angle wasn't measured from the correct reference line.
- Forgetting that a force perpendicular to a chosen axis contributes zero to that axis, and including it anyway.
- Missing a force entirely from the diagram (most often a reaction force or the weight).
- Not stating a positive direction before assigning signs, leading to sign errors when combining equations.

6.2 Coefficient of Friction and Limiting Equilibrium

Concept

When a particle rests on a **rough** surface (one that is not smooth), a **frictional force** F can act along the surface, opposing any tendency of the particle to slide. Friction is a *reactive* force: it takes whatever value is needed to prevent sliding, up to a maximum.

That maximum is set by the **coefficient of friction** μ (a property of the two surfaces in contact, with no units) and the **normal reaction** R between the surfaces:

$$F \leq \mu R \quad \text{at all times}$$

When the particle is on the very point of sliding — **limiting equilibrium** — friction takes its maximum possible value, $F = \mu R$, and this same value is used once the particle is actually sliding. (A Level mechanics does not require distinguishing separate static and kinetic coefficients; one μ is used throughout unless a question states otherwise.) A **smooth** surface is the special case $\mu = 0$, so $F = 0$ always.

Key vocabulary

Term	Meaning
Coefficient of friction (μ)	A number ($0 \leq \mu$, typically < 1) describing how rough two surfaces are; no units
Normal reaction (R)	The contact force perpendicular to the surface, pushing the particle away from it
Rough	A surface that can exert friction ($\mu > 0$)
Limiting equilibrium	The particle is in equilibrium but on the point of sliding; friction is at its maximum value
Limiting friction	The maximum possible frictional force, $F = \mu R$
Dynamic (kinetic) friction	Friction acting on a particle already sliding; taken equal to μR at A Level

Key formulas

KEY FORMULA

$$F \leq \mu R$$

$$F = \mu R \quad (\text{limiting equilibrium, or once sliding})$$

Procedure

1. Draw the force diagram, including the normal reaction R and the frictional force F acting along the surface, in the direction that opposes the particle's actual or tendency-to motion.
2. Resolve perpendicular to the surface to find R (often simply $R = mg$ on a horizontal surface with no other vertical forces).
3. Decide whether the particle is in general equilibrium ($F \leq \mu R$, F unknown but bounded) or in limiting equilibrium / already moving ($F = \mu R$ exactly). Never assume $F = \mu R$ unless the question states or implies the particle is on the point of, or already, sliding.
4. Resolve along the surface and solve for the unknown, substituting $F = \mu R$ only where justified in step 3.

Worked example

EXAMPLE 6.2

Question. A block of mass 8 kg rests on a rough horizontal surface. The coefficient of friction

between the block and the surface is 0.4. A horizontal force P is applied to the block. (a) Find the value of P at which the block is on the point of sliding. (b) State, with a reason, whether the block moves if $P = 35 \text{ N}$.

Step 1 (part a) — resolve vertically to find the normal reaction. M1

$$R = mg = 8 \times 9.8 = 78.4 \text{ N} \quad \text{A1}$$

Step 2 (part a) — the block is on the point of sliding, so it is in limiting equilibrium and $F = \mu R$. M1

$$F = \mu R = 0.4 \times 78.4 = 31.36 \text{ N}$$

Step 3 (part a) — since the block is in equilibrium at this point, the applied force must exactly balance the limiting friction. A1

$$P = F = 31.36 \text{ N}$$

Step 4 (part b) — compare the applied force with the maximum possible friction force. M1

Maximum available friction = 31.36 N . Since $35 \text{ N} > 31.36 \text{ N}$, the applied force exceeds the maximum friction that the surface can provide, so the block cannot remain in equilibrium — it moves. **A1**

Answer: (a) $P = 31.36 \text{ N}$ ($\approx 31.4 \text{ N}$). (b) The block moves, since the applied force of 35 N exceeds the limiting friction of 31.36 N .

Exam tip

EXAM TIP

What this idea builds on: it is tempting to apply $F = \mu R$ as a reflex whenever friction appears in a question, but this substitution is only valid at limiting equilibrium or once the particle is actually sliding. Many questions instead describe a *general* equilibrium scenario — a block simply resting, or held in place by a force too small to move it — where $F < \mu R$, and F must be found by resolving rather than by substituting μR directly. Using $F = \mu R$ in this situation is one of the most common errors at this level and scores no marks. Get into the habit of asking "does this question say the particle is limiting, on the point of moving, or already sliding?" before writing $F = \mu R$.

Common exam mistakes

- Using $F = \mu R$ automatically, even when the particle is in general (non-limiting) equilibrium — F must instead be found by resolving.
- Calculating R using mg alone when another force (e.g. an applied force with a vertical component) also contributes to the vertical equilibrium.
- Giving friction the wrong direction on the diagram — it always opposes the direction of sliding or impending sliding, never the direction of an applied force by default.
- Confusing μ (a pure number) with a force and giving it units of N .

6.3 Bodies on Inclined Planes

Concept

On an inclined plane, the sensible axes are no longer horizontal and vertical but **parallel** and **perpendicular** to the slope, because the normal reaction and any motion both align with those directions.

The particle's weight mg must therefore be resolved into a component $mg \sin \alpha$ *down the slope* (parallel) and a component $mg \cos \alpha$ *into the slope* (perpendicular), where α is the angle of inclination to the horizontal.

A helpful way to remember which is which: as the angle α gets smaller, the plane gets flatter, so the component *into* the slope should get closer to the full weight mg — and $\cos \alpha \rightarrow 1$ as $\alpha \rightarrow 0$, which matches the perpendicular component $mg \cos \alpha$. The component *along* the slope should shrink to zero on a flat plane, matching $mg \sin \alpha$, since $\sin \alpha \rightarrow 0$ as $\alpha \rightarrow 0$.

Perpendicular to the plane there is no acceleration — the particle cannot move through the solid plane — so R is found first, from $R = mg \cos \alpha$ (adjusted for any other perpendicular forces). Along the plane, friction (if the surface is rough) acts up or down the slope depending on which way the particle is sliding or tends to slide, and Newton's second law is applied exactly as on a horizontal surface, just along the tilted axis.

Key vocabulary

Term	Meaning
Angle of inclination (α)	The angle the plane makes with the horizontal
Component of weight	$mg \sin \alpha$ acting down the slope, $mg \cos \alpha$ acting into the slope
Parallel / perpendicular to the plane	The two axes used for resolving on an incline, replacing horizontal/vertical
Smooth incline	$\mu = 0$ on the slope; no frictional force
Rough incline	$\mu > 0$ on the slope; friction opposes sliding or impending sliding

Key formulas

KEY FORMULA

$$R = mg \cos \alpha$$

$$a = g \sin \alpha \quad (\text{smooth incline, sliding freely down})$$

$$mg \sin \alpha - F = ma \quad (\text{rough incline, sliding down; friction acts up the slope})$$

where, for a rough incline, $F = \mu R = \mu mg \cos \alpha$ once the particle is sliding or in limiting equilibrium.

Procedure

1. Draw the incline and mark the angle α , the weight mg (vertically down), the normal reaction R (perpendicular to the plane), and, if rough, the friction F (along the plane, opposing sliding/impending sliding).
2. Resolve perpendicular to the plane to find R — this direction has zero acceleration, whatever the particle is doing along the plane.
3. If the surface is rough and the particle is sliding or in limiting equilibrium, calculate $F = \mu R$; otherwise leave F as an unknown bounded by $F \leq \mu R$.
4. Resolve parallel to the plane, choosing a positive direction (usually the direction of actual or expected motion), and apply equilibrium (sum = 0) or Newton's second law (sum = ma) to solve for the required quantity.

Worked example

EXAMPLE 6.3

Question. A particle of mass 3 kg is released from rest on a rough plane inclined at 25° to the horizontal. The coefficient of friction between the particle and the plane is 0.3. Find the acceleration of the particle as it slides down the plane.

Step 1 — resolve perpendicular to the plane to find R . M1

$$R = mg \cos 25^\circ = 3 \times 9.8 \times \cos 25^\circ = 26.6 \text{ N (3 s.f.)} \quad \text{A1}$$

Step 2 — the particle is sliding down, so friction acts up the plane at its dynamic value $F = \mu R$. M1

$$F = \mu R = 0.3 \times 26.6 = 7.99 \text{ N} \quad \text{A1}$$

Step 3 — apply Newton's second law along the plane, taking down the slope as positive. M1

$$mg \sin 25^\circ - F = ma$$

Step 4 — substitute the known values and simplify.

$$(3)(9.8) \sin 25^\circ - 7.99 = 3a \Rightarrow 12.42 - 7.99 = 3a \Rightarrow 4.43 = 3a \quad \text{A1}$$

Step 5 — solve for a .

$$a = 4.43 \div 3 = 1.48 \text{ m s}^{-2} \text{ (3 s.f.)}$$

Answer: $a = 1.48 \text{ m s}^{-2}$ (3 s.f.), directed down the plane.

Exam tip

EXAM TIP

Why the reasoning matters, not just the answer: it is not enough to write $R = mg \cos \alpha$ from memory — this equation is only true *because* there is no acceleration perpendicular to the plane, and mark schemes often award a mark specifically for stating that reasoning before the equation is used. It is also worth building in a sanity check for the sin/cos split: since $g \sin \alpha$ and $g \cos \alpha$ are very easy to swap under exam pressure, checking that as $\alpha \rightarrow 0$ (the plane becomes flat) the perpendicular component should tend to mg — i.e. $\cos \alpha \rightarrow 1$ — catches this error reliably.

Common exam mistakes

- Swapping $\sin \alpha$ and $\cos \alpha$ when resolving the weight. A useful check: the component *along* the slope uses $\sin \alpha$, the component *into* the slope uses $\cos \alpha$.
- Forgetting that $R \neq mg$ once the plane is tilted — using $R = mg$ instead of $R = mg \cos \alpha$.
- Getting the direction of friction wrong when a particle is sliding *up* the plane (decelerating): friction still opposes the actual direction of motion, so it acts *down* the slope in that case, adding to the deceleration.
- Sign errors when a particle decelerates up a slope versus accelerates down it — always state the positive direction chosen before writing the equation of motion.

6.4 Equilibrium of Connected Particles with Friction

Concept

This section combines the connected-particle and pulley ideas of Chapter 3 with friction and inclined planes — a classic synthesis question that pulls together most of the chapter.

A particle A sits on a rough inclined plane, connected by a light inextensible string, passing over a smooth pulley fixed at the top of the plane, to a particle B hanging freely. Because the string is inextensible and the pulley smooth:

- both particles move with the same magnitude of acceleration a , and
- the tension T is the same throughout the string.

Each particle needs its own equation of motion, written in its own direction of positive motion (down the slope or up the slope for A; downward for B, since B falls as A is pulled up, or vice versa). The two equations are then solved simultaneously, usually by adding them together to eliminate T .

Key vocabulary

Term	Meaning
Connected particles	Two or more particles joined by a string (or rod) so their motions are linked
Common acceleration	The single value of a shared by both particles, because the string is inextensible
Tension (T)	The force the string exerts, pulling equally on both particles it connects
Smooth pulley	A pulley exerting no friction, so it only changes the direction of the tension, not its magnitude
Taut string	An assumption that the string remains straight and under tension throughout the motion

Key formulas

KEY FORMULA

$$T - m_A g \sin \alpha - F = m_A a \quad (\text{particle A on the incline, moving up the slope})$$

$$m_B g - T = m_B a \quad (\text{particle B, hanging, moving downward})$$

with $F = \mu R = \mu m_A g \cos \alpha$ once A is sliding or in limiting equilibrium. Adding the two equations eliminates T and gives a directly; substituting a back into either equation then gives T .

Procedure

1. Sketch both particles separately, marking every force on each (weight, tension, and for the particle on the incline, R and F).
2. State the direction each particle actually moves in (from the context, or from whichever mass "wins"), and use this to fix the direction of friction on the particle on the incline.
3. For the particle on the incline: resolve perpendicular to find R , calculate $F = \mu R$, then write the equation of motion parallel to the plane, including T .
4. For the hanging particle: write its equation of motion directly (weight and tension are already vertical).
5. Solve the two equations simultaneously — usually by adding them so T cancels — to find a , then substitute back to find T .

Worked example

EXAMPLE 6.4

Question. Particles A (mass 2 kg) and B (mass 5 kg) are connected by a light inextensible string passing over a smooth pulley fixed at the top of a rough plane inclined at 30° to the horizontal. A lies on the inclined plane; B hangs freely below the pulley. The coefficient of friction between A and the plane is 0.25. The system is released from rest, with B moving downward. Find (a) the acceleration of the system, (b) the tension in the string.

Step 1 — establish the direction of friction, then resolve perpendicular to the plane for A. As B falls, A is pulled up the slope, so friction on A acts down the slope, opposing its motion. **M1**

$$R = m_A g \cos 30^\circ = 2 \times 9.8 \times \cos 30^\circ = 17.0 \text{ N (3 s.f.)} \quad \text{A1}$$

Step 2 — A is sliding, so $F = \mu R$. **M1**

$$F = 0.25 \times 17.0 = 4.24 \text{ N} \quad \text{A1}$$

Step 3 — write the equation of motion for A along the plane, taking up the slope (its direction of motion) as positive. **M1**

$$T - m_A g \sin 30^\circ - F = m_A a \Rightarrow T - 9.8 - 4.24 = 2a$$

Step 4 — write the equation of motion for B, taking downward (its direction of motion) as positive. **M1**

$$m_B g - T = m_B a \Rightarrow 49 - T = 5a$$

Step 5 — add the two equations to eliminate T . Rewriting A's equation as $T = 9.8 + 4.24 + 2a$ and substituting into B's equation: **M1**

$$49 - (9.8 + 4.24 + 2a) = 5a \Rightarrow 49 - 9.8 - 4.24 = 7a$$

Step 6 — solve for a.

$$34.96 = 7a \Rightarrow a = 4.99 \text{ m s}^{-2} \text{ (3 s.f.)}$$

Step 7 — substitute a back into either equation to find T. Using B's equation: **M1**

$$T = m_B g - m_B a = 49 - 5(4.99) = 24.0 \text{ N (3 s.f.)} \quad \textbf{A1}$$

Answer: (a) $a \approx 4.99 \text{ m s}^{-2}$. (b) $T \approx 24.0 \text{ N}$.

Exam tip

EXAM TIP

Why this feels like a big step up: this section asks you to combine several ideas at once — connected particles, a pulley, friction and an incline — into a single question, which is a significant jump in complexity from handling each idea on its own. The key discipline that makes it manageable is treating each particle *completely separately*: its own diagram, its own positive direction, its own equation — before combining them. If you try to reason about the "whole system" informally, you very often get a sign wrong on the friction or weight-component term. Once a is found, this type of question is often followed by a kinematics part using the suvat equations (linking back to Chapter 2).

Common exam mistakes

- Using the same positive direction convention for A and B without checking it correctly matches how each particle actually moves (up-the-slope for A must correspond to downward for B, not upward).
- Forgetting friction altogether when a rough incline is combined with a pulley, treating it as if the plane were smooth.
- Getting R wrong for the particle on the incline by using mg instead of $mg \cos \alpha$, which then gives the wrong F.
- Sign errors when adding the two equations of motion to eliminate T — write both equations with T on the same side before adding.

Chapter summary

Idea	Key point
Resolving at an angle	$F \cos \theta$ along an axis, $F \sin \theta$ perpendicular to it; equilibrium needs both sums to be zero
Friction	$F \leq \mu R$ in general; $F = \mu R$ only at limiting equilibrium or once sliding
Inclined planes	Resolve parallel/perpendicular to the slope: $R = mg \cos \alpha$, and $mg \sin \alpha - F = ma$ along the slope
Connected particles with friction	Write a separate equation of motion for each particle, in its own direction of motion, then solve simultaneously to eliminate T

Common exam mistakes

- Applying $F = \mu R$ when the particle is not at limiting equilibrium or actually sliding.
- Swapping $\sin \alpha$ and $\cos \alpha$ when resolving weight on an incline.
- Inconsistent positive-direction conventions between two connected particles.
- Omitting a force (most often friction or the normal reaction) from a force diagram before resolving.

You should now be able to

- Resolve a force into two perpendicular components at any angle, and choose sensible axes for a given problem.
- Apply $F \leq \mu R$ and $F = \mu R$ correctly, distinguishing general equilibrium from limiting equilibrium or motion.
- Solve problems involving a particle in equilibrium or accelerating on a smooth or rough inclined plane, resolving parallel and perpendicular to the slope.
- Analyse connected particles joined by a string over a smooth pulley, including a particle on a rough inclined plane, by writing and solving simultaneous equations of motion.